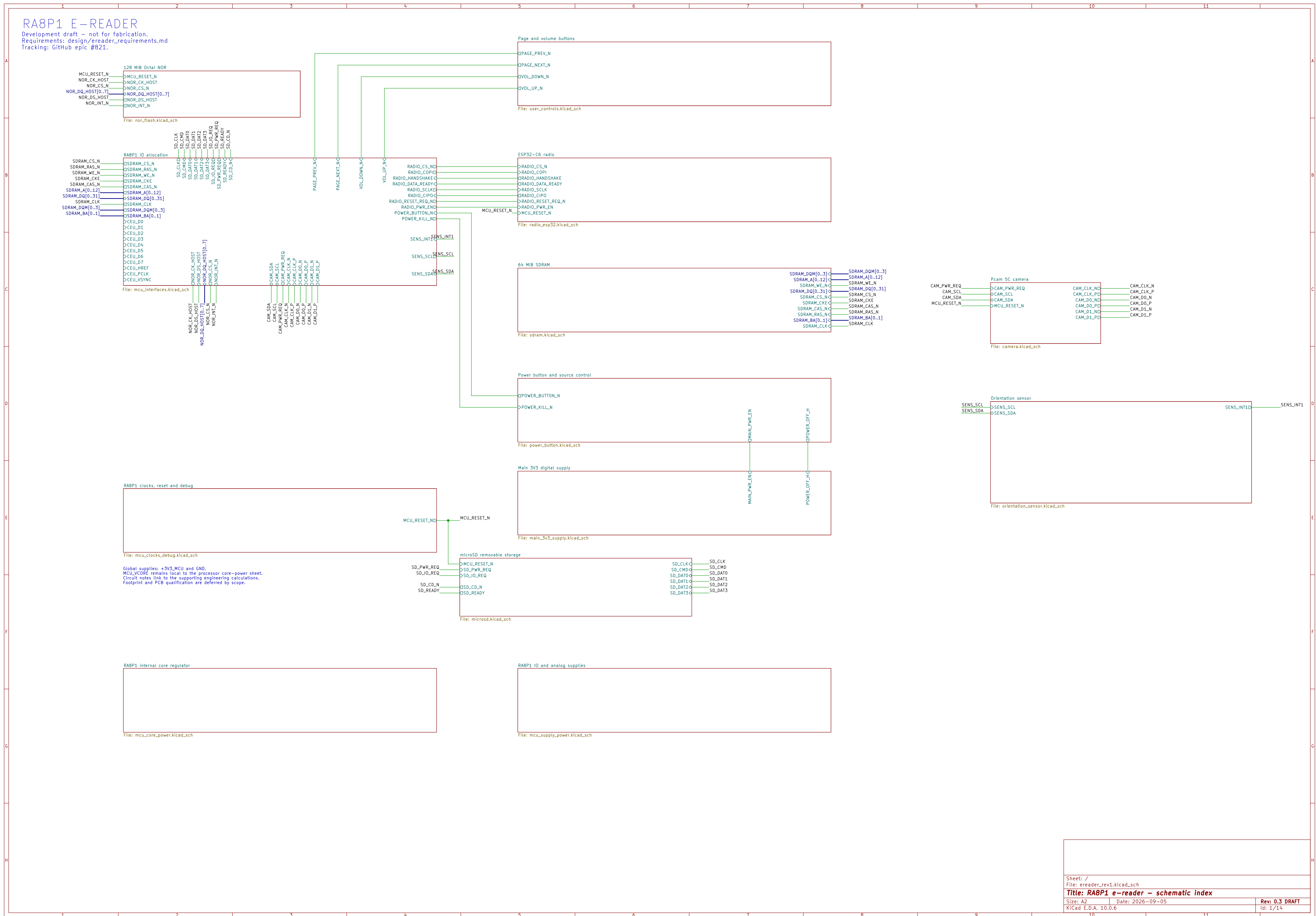


Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.



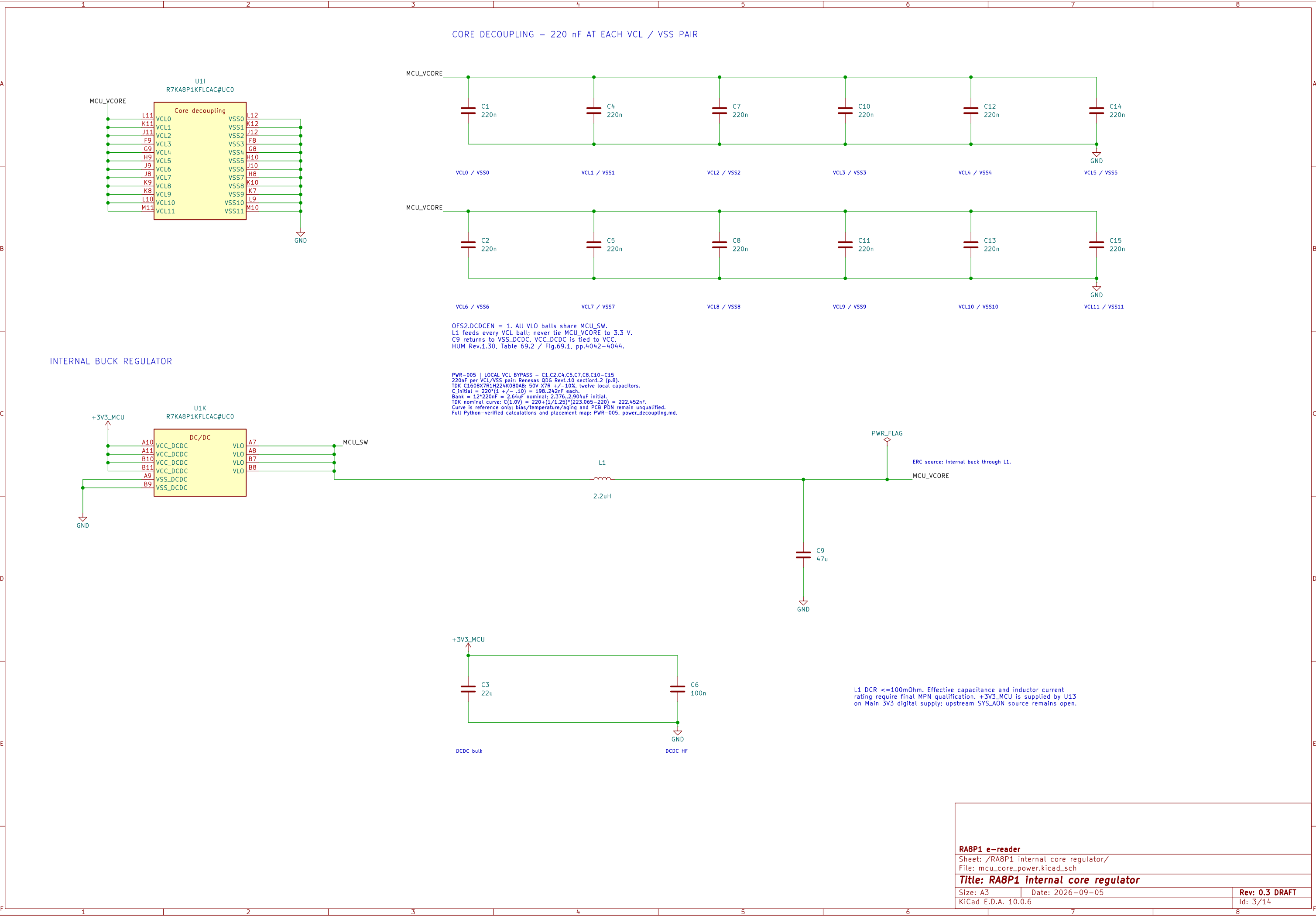
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

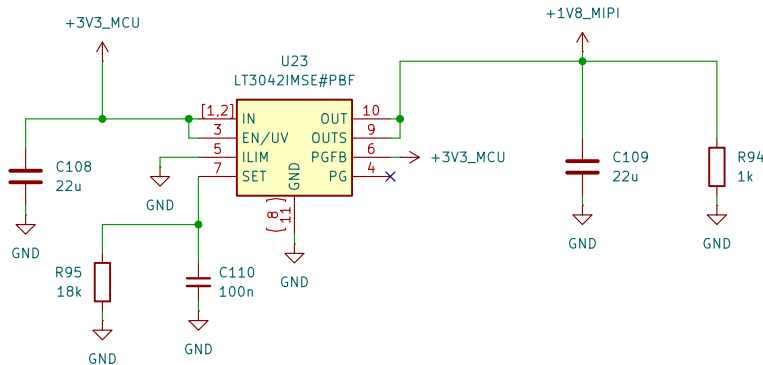
The diagram illustrates the IO allocation for the RABP1 e-reader. It includes several key components and their connections:

- Power and Grounding:** +3V3_MCU and +3V3_USBHS_A are connected to various components. A 120R resistor (FB1) is used for the +3V3_MCU supply. A 10uF capacitor (C63) is connected to the +3V3_USBHS_A supply.
- USB Interface:** The USB-004 module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 10uF capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- SDRAM:** The SDRAM module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 10uF capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- GPIOs:** The GPIO module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 10uF capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- Other Components:** The diagram also shows connections for the +3V3_MCU, +3V3_USBHS_A, and various other components like the +3V3_MCU, +3V3_USBHS_A, and various other components.

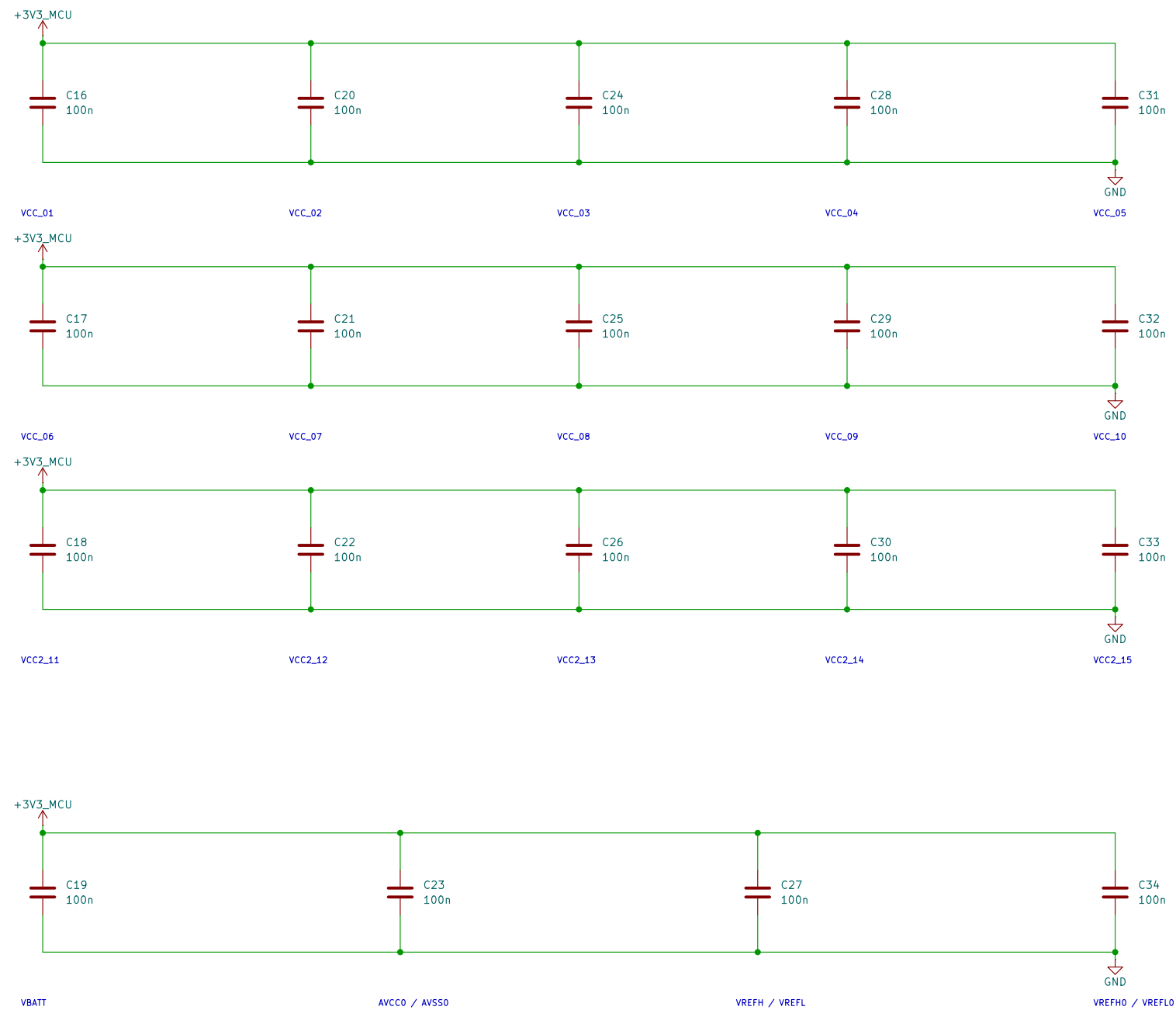
The diagram is a detailed schematic showing the electrical connections between the RABP1 e-reader and its various components. It includes a title block with the following information:

- Title:** RABP1 IO allocation
- Sheet:** /RABP1 IO allocation/
- File:** mcu_interfaces.kicad_sch
- Size:** A2
- Date:** 2026-09-05
- Rev:** 0.3 DRAFT
- Id:** 2/14



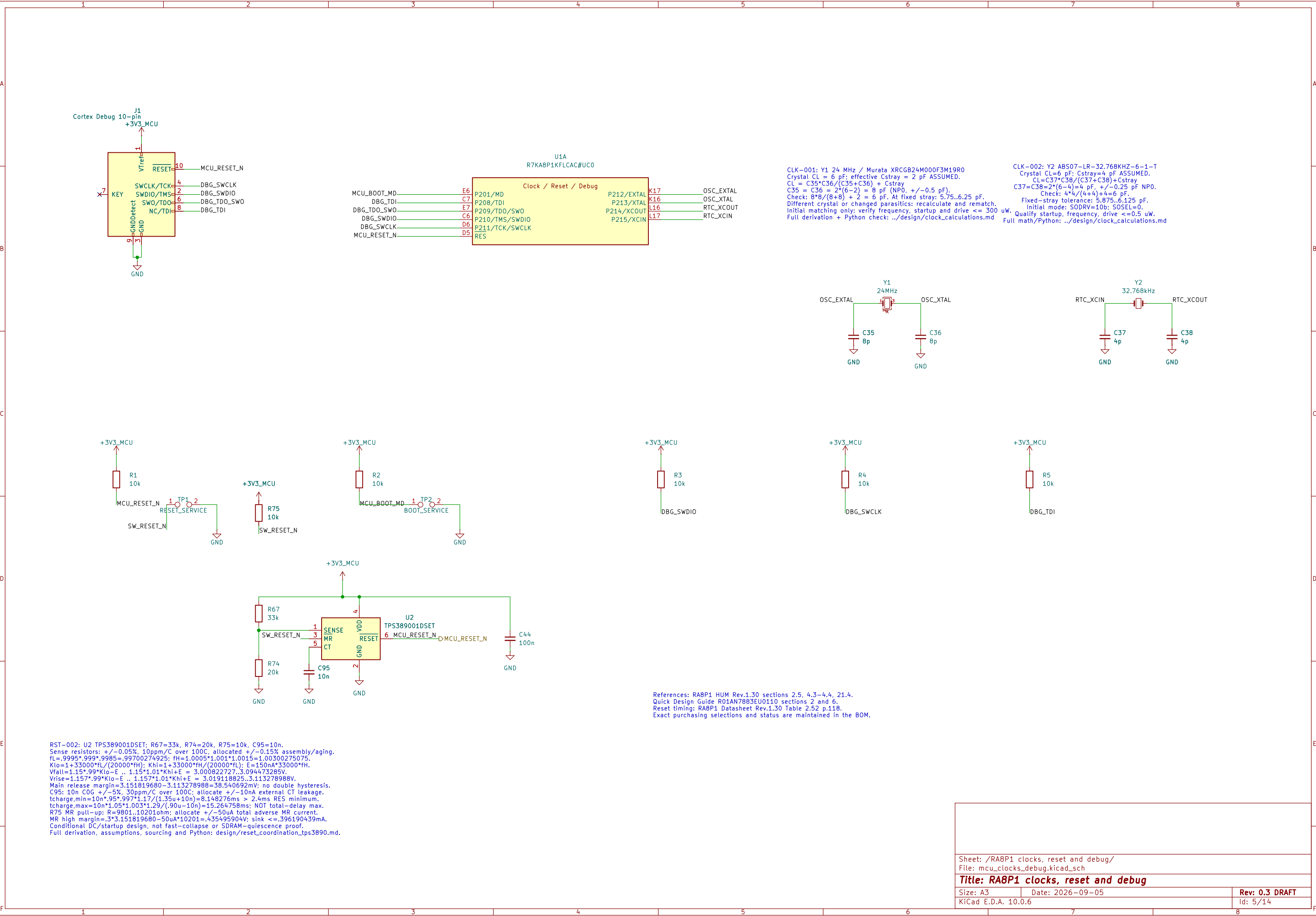


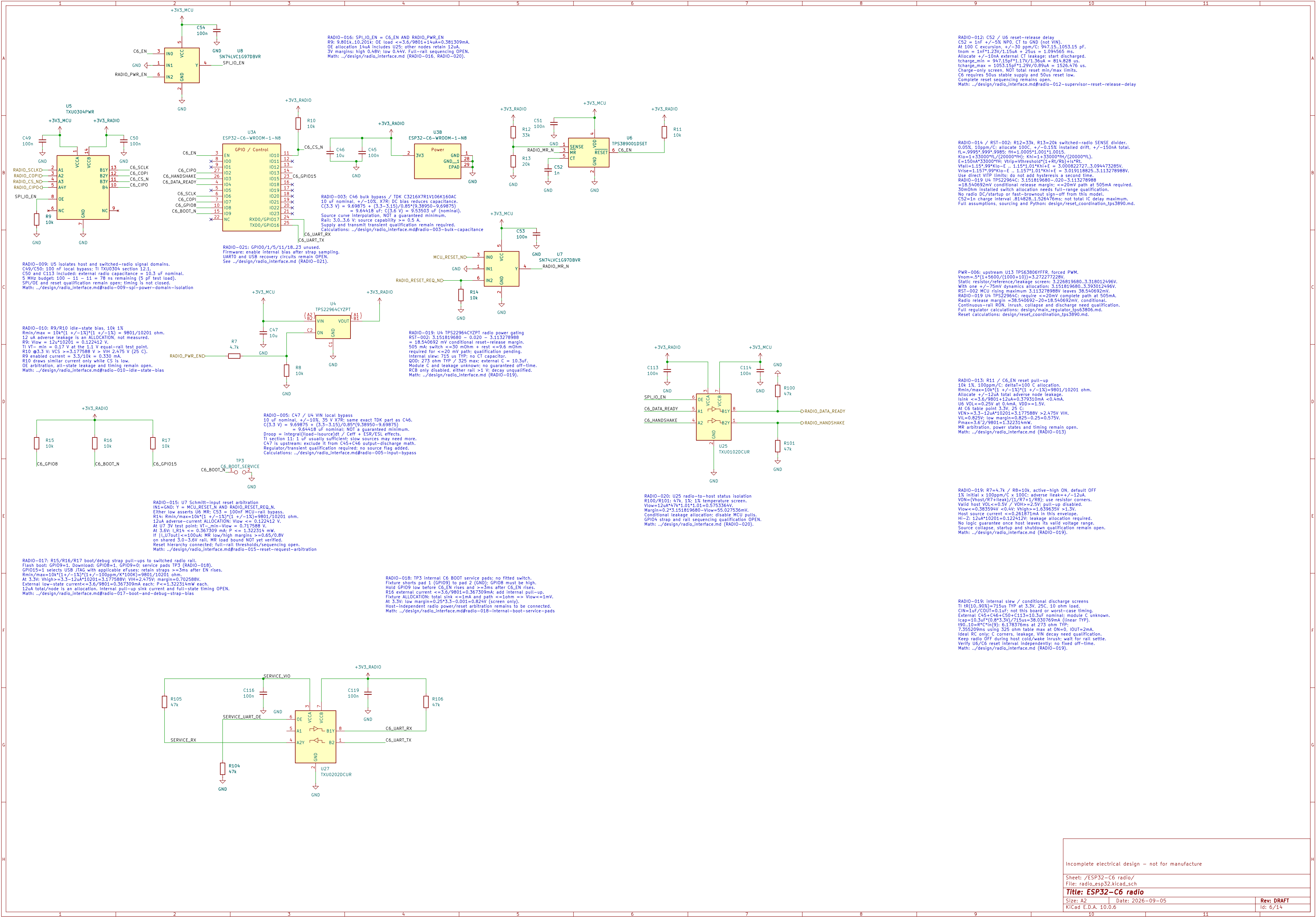
I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

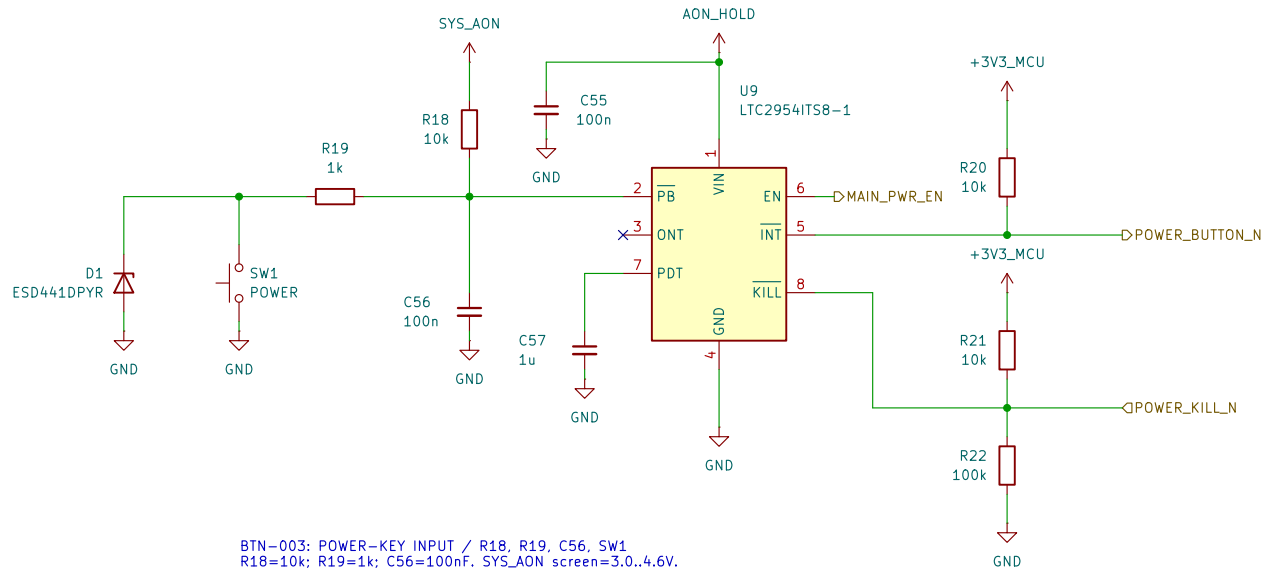


CMS-013: +20mA steady CSI branch planning allocation
4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
+ 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
13mA is an engineering envelope, not an interpolated guarantee.
Auxiliary allowance requires verification: startup/inrush excluded.
AVCC_MIP1 remains within existing MCU allocation.
Main subunit: 2.075 + 0.020 = 2.095A; full system budget pending.
At 2.0496V, 3.2V input, 1.25W average, 75% efficiency:
input 2.961817158A; converter loss = 2.3694535727W
Source, current and thermal acceptance remain open.
Sources, conditions and Python: design/camera_storage/interfaces.md.

RA8P1 e-reader		
Sheet: /RA8P1 IO and analog supplies/ File: mcu_supply_power.kicad_sch		
Title: RA8P1 I/O and analog supplies		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6		Id: 4/14

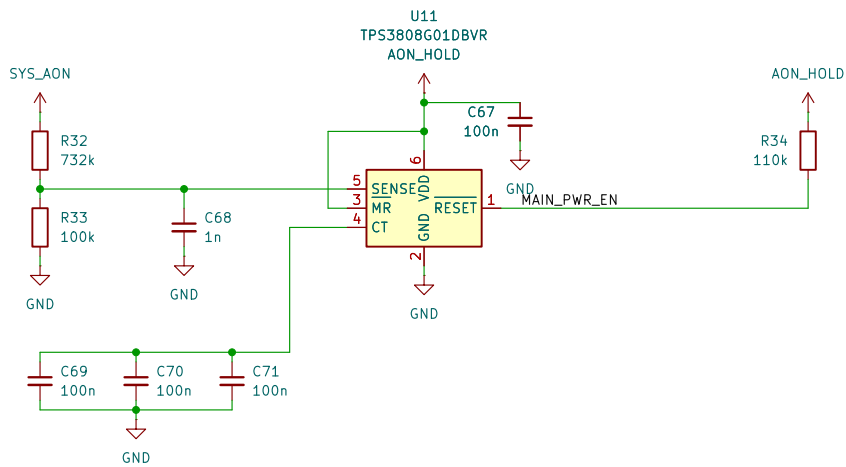






BTN-003: POWER-KEY INPUT / R18, R19, C56, SW1
R18=10k; R19=1k; C56=100nF. SYS_AON screen=3.0..4.6V.
R bounds: nominal*(0.99*0.99)..nominal*(1.01*1.01).
At PB=0.6V: Isource<=(4.6-0.6)/9801 + 15uA + 12uA
<435.13uA; Isink=0.6/(1020.1+0.5)>587.88uA.
Thus the closed switch pulls PB below its 0.6V falling threshold.
Min contact current=3/(10201+1020.1+0.5)-12uA>255.34uA.
Open contact>=2.87748V; SW1 minimum rating=10uA at 2V.
C56 discharge peak<=4.6/980.1<4.694mA < SW1 50mA rating.
12uA is a leakage ALLOCATION; ESD/transient validation required.
Full derivation + Python: ../design/single_button_power.md BTN-003/005/010.

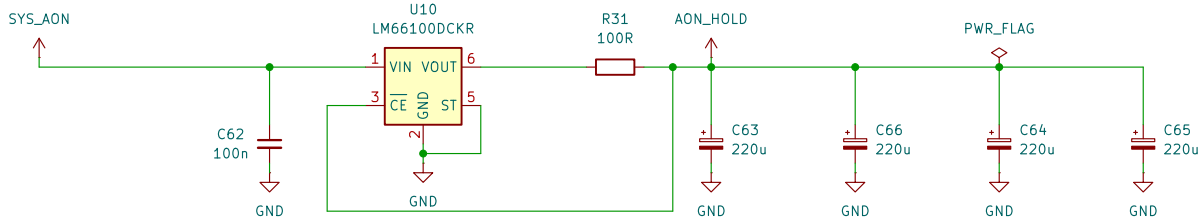
BTN-004: HARDWARE FORCED-OFF TIMING / C57
C57=1uF; TDK C2012X7R1E105K125AB, 25V X7R +/-10%.
C[uF]=1.56e-4*(t_added[ms]-1).
t_force,nom=64ms + 1ms + 1/1.56e-4=6475.256ms (6.48s).
C tolerance/temp only: 1*0.90*0.85=0.765uF.
1*1.10*1.15=1.265uF -> 4.969..8.174s using NOMINAL IC equation.
Not guaranteed timing: excludes IC spread, DC bias, aging, leakage.
ONT intentionally open: internal turn-on debounce=26..41ms.
Release the initial ON press; after forced off, release then press again.
Full calculation + Python: ../design/single_button_power.md BTN-004/005.



SYS-007 / SENS-010: SHUTDOWN / U12, Q1-Q3, R35-R40, R70-R71
POWER_OFF_H=NOT(MAIN_PWR_EN); held-domain VCC=2.7..4.6V.
Each gate branch: 1k series / 1M pulldown; 1% +100ppm/C, 100C.
4-gate peak<=4*4.6/(1000*0.99^2)=18.773595mA <50mA.
Includes future USB-clear; static<=22.773595uA <100uA.
VGSmin=(2.6-1020.1*1uA)/(1+1020.1/980100)=2.596278V.
1uA per gate/board leakage (is a qualification allocation).
RUN/KILL/discharge/USB-clear FET drains remain separate.
R39||R40 + Q2: Rmax=22*1.01^2/2+0.2=11.4211ohm.
Irc<=2.5mA: 1.5mA key-filter return +1mA other injection.
Vinf=Irc*Rmax=28.552750mV; Cmain<=1mF.
Total=10ms+Rmax*Cmain*ln((3.6-Vinf)/(0.09-Vinf))+10ms
=66.398757ms <75.794309ms hold; margin9.395552ms.
Each 22R: P<=3.6^2/(22*0.99^2)=0.601052W; 1W at70C.
Qualify FET resistance, response, derating, backfeed and cycling.
Gate/transient charge shares 1uC reserve; allocations need validation.
VBATT firmware/option invariant remains unimplemented.
Sources/Python: ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

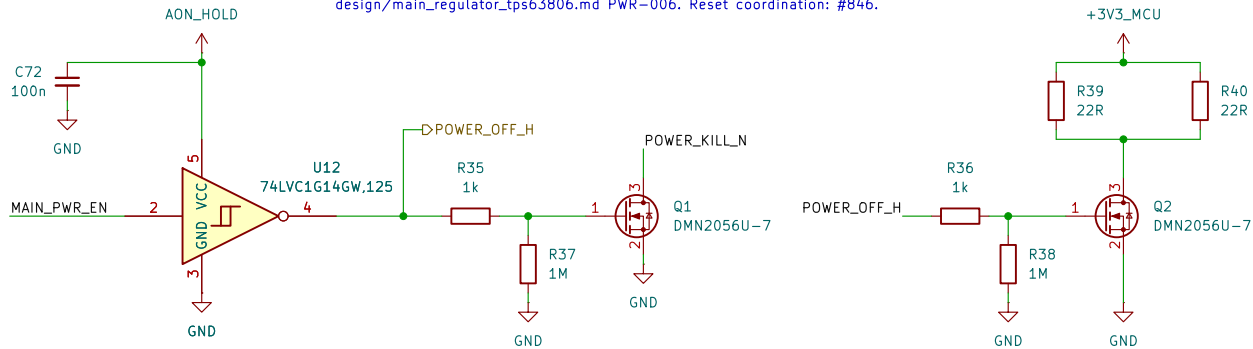
INCOMPLETE POWER CONTROL - NOT FOR FABRICATION
PB, INT, KILL, held supply and source supervisor are connected.
U12/Q1/Q2 provide fault KILL clamp and main-rail discharge.
MAIN_PWR_EN feeds raw U16; held Q3 separately clamps U13 EN.
SYS_AON source and USB permission-latch clear remain to be built.
Current converter: U13 TPS63806; design/main_regulator_tps63806.md PWR-006.
MCU reset: design/reset_coordination_tps3890.md RST-002.
Radio release margin and system shutdown coordination remain open (#846).

ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.



SYS-007 / SENS-010: HELD SUPPLY / U10, R31, C62-C66
C63-C66: 4 x 220uF/10V; Cmin=880uF*0.9^3=641.52uF.
VIN=raw SYS; VOUT->R31->reservoir; CE senses reservoir.
Icontrol<=1.825mA incl 1.100mA bank leakage screen.
Reverse allowance=0.817mA; Itotal<=2.642mA.
Vinitial=3.263705831-max(0.250,1.825mA*102.24ohm)
=3.013705831V; shared transition reserve=1uC.
thold=[641.52uF*(Vinitial-2.7V)-1uC]/2.642mA
=75.794309ms, conditional allocation.
Main C<=1mF. Rdis<=11.4211ohm, return/injection<=2.5mA;
10ms response +46.398757ms to90mV +10ms low hold
=66.398757ms total; hold margin=9.395552ms.
Recovery: raw>=3.70V, held>=3.25V; 1s settling PLUS
full source qualification; recharge model=0.657961s.
Qualify startup leakage, raw VIN=0, backfeed and cycling.
See ../design/sensor_alternative.md SENS-010;
../design/system_power_design.md SYS-007 for topology.

SYS-007 / PWR-006: SOURCE SUPERVISION / U11, R32-R34, C67-C71
Vtrip=Vref*(1+R32/R33)+Isense*R32.
Nominal=0.405*(1+732k/100k)=3.369600V.
Vref +/-2%, Isense +/-25nA; each R +/-0.1%, 25ppm/C, 100C.
Corner trip=3.263705831..3.476597632V.
C68: tau=(732k||100k)*1n=87.980769us (nominal).
C69-C71: 3x100n C0G; initial/temp=284.145..316.827nF.
CT delay=300/175+0.0005=1.714786s nominal.
0.6*(284.145/175+0.0005)=0.974511s MODEL, not guarantee.
Require qualified reset hold >=0.90s >650ms KILL blank +10ms.
R34=110k; ENhigh>=2.7-3.8uA*110k*1.01^2=2.273598V.
POR sink=(1.3-0.2)/(110k*0.99^2)+3.8uA=14.003041uA.
MAIN_PWR_EN drives raw U16 buffer; held Q3 clamps TPS63806 EN.
Held U12 also drives KILL/discharge; USB-clear still needed.
Current EN network and rail calculations: PWR-006.
Sources/Python: design/system_power_design.md SYS-007;
design/main_regulator_tps63806.md PWR-006. Reset coordination: #846.



Sheet: /Power button and source control/
File: power_button.kicad_sch

Title: RABP1 E-reader - power button and source control

Size: A3

Date: 2026-09-07

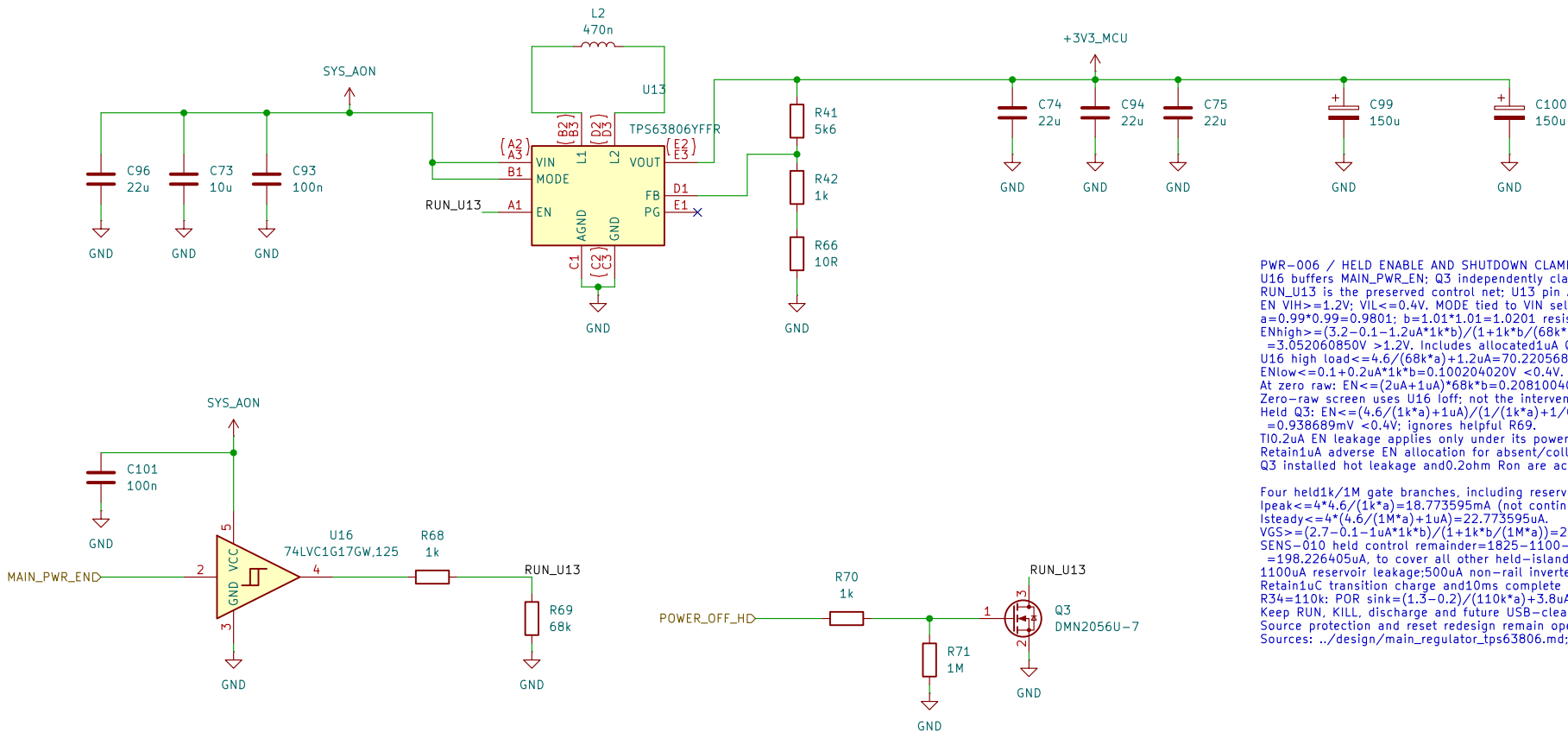
Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 7/14



MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



PWR-006 / HELD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWR_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
a=0.99*0.99=0.9801; b=1.01*1.01=1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b/(68k*a))
=3.052060850V >1.2V. Includes allocated1uA Q3 off leakage.
U16 high load<=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
TI0.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Ron are acceptance allocations.

Four held1k/1M gate branches, including reserved USB-clear:
lpeak<=4*4.6/(1k*a)=18.773595mA (not continuous rating).
lsteady<=4*(4.6/(1M*a)+1uA)=22.773595uA.
VGS>=(2.7-0.1-1uA*1k*b)/(1+1k*b/(1M*a))=2.596278V.
SENS-010 held control remainder=1825-1100-500-4-22.773595
=198.226405uA, to cover all other held-island loads.
1100uA reservoir leakage;500uA non-rail inverter-input allocation.
Retain1uC transition charge and10ms complete response allocation.
R34=110k; POR sink=(1.3-0.2)/(110k*a)+3.8uA=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Sources: ../design/main_regulator_tps63806.md; SENS-010 revision.

PWR-006 / FEEDBACK AND LOCAL PASSIVES
Vnom=0.500*(1+5600/(1000+10))=3.272277228 V.
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed bounds: initial*TCR(100C)*aging*assembly, then ohms.
R41=5600*(1+/-0.0002)*(1+/-0.0005)*(1+/-0.001)
*(1+/-0.0005)+/-0.02 = 5587.669237..5612.349243 ohm.
R42=1000*(1+/-0.0001)*(1+/-0.0002)*(1+/-0.001)
*(1+/-0.0005)+/-0.01 = 998.190970..1001.810970 ohm.
R66=10*(1+/-0.01)*(1+/-0.02)*(1+/-0.01)+/-0.05
=9.554980..10.455020 ohm; one combined aging/assembly 1%.
Vlo=0.495*(1+R41min/(R42max+R66max))-100nA*R41max.
Vhi=0.505*(1+R41max/(R42min+R66min))+100nA*R41max.
Vstatic=3.226819680..3.318012496 V (FPWM only).
One combined +/-75mV regulation/ripple/transient/routing allowance:
Vcomplete=3.151819680..3.393012496 V; qualification required.
SDRAM static VOH margin=2.4-0.7*3.393012496=24.891253mV.
This is not measured SI/noise headroom or reset qualification.

L2: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
=>0.3708..0.5698uH versus TI0.37..0.57uH. Qualify hot/current.
C74/C75/C94: 3*22uF*0.8*0.85*0.6=26.928uF > TI21uF.
C96: 22uF*0.8*0.85*0.6=8.976uF > TI4uF input minimum.
0.6 is allocated combined bias/aging retention. NOT guaranteed.
Samsung typical output-bias retention at3.393012496V=90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LOAD, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold-start screen, not measured load.
Pout=3.393012496*1.8=6.107422493W.
At allocated75% efficiency: Pin=8.143229991W; lin@3.2V=2.544759372A.
Loss=2.035807498W: NOT an acceptable continuous thermal result.
At90% efficiency: loss=0.678602499W.
TJ screen=70C+78.8C/W*0.678602499W=123.474C.
TI board theta-JA is not the final PCB thermal resistance.
2.25A screen: Pin=3.393012496*2.25/0.75=10.179037489W.
Converter limit, source limit, startup and thermal envelope need validation.

PWR-006 storage checkpoint (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on adds10.2uF =>489.81uF.
C99/C100 remain150uF each, C94 adds22uF to the output.
Old bootstrap/private VCC/VC capacitors were not main-rail storage.
Whole-main ceiling remains1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

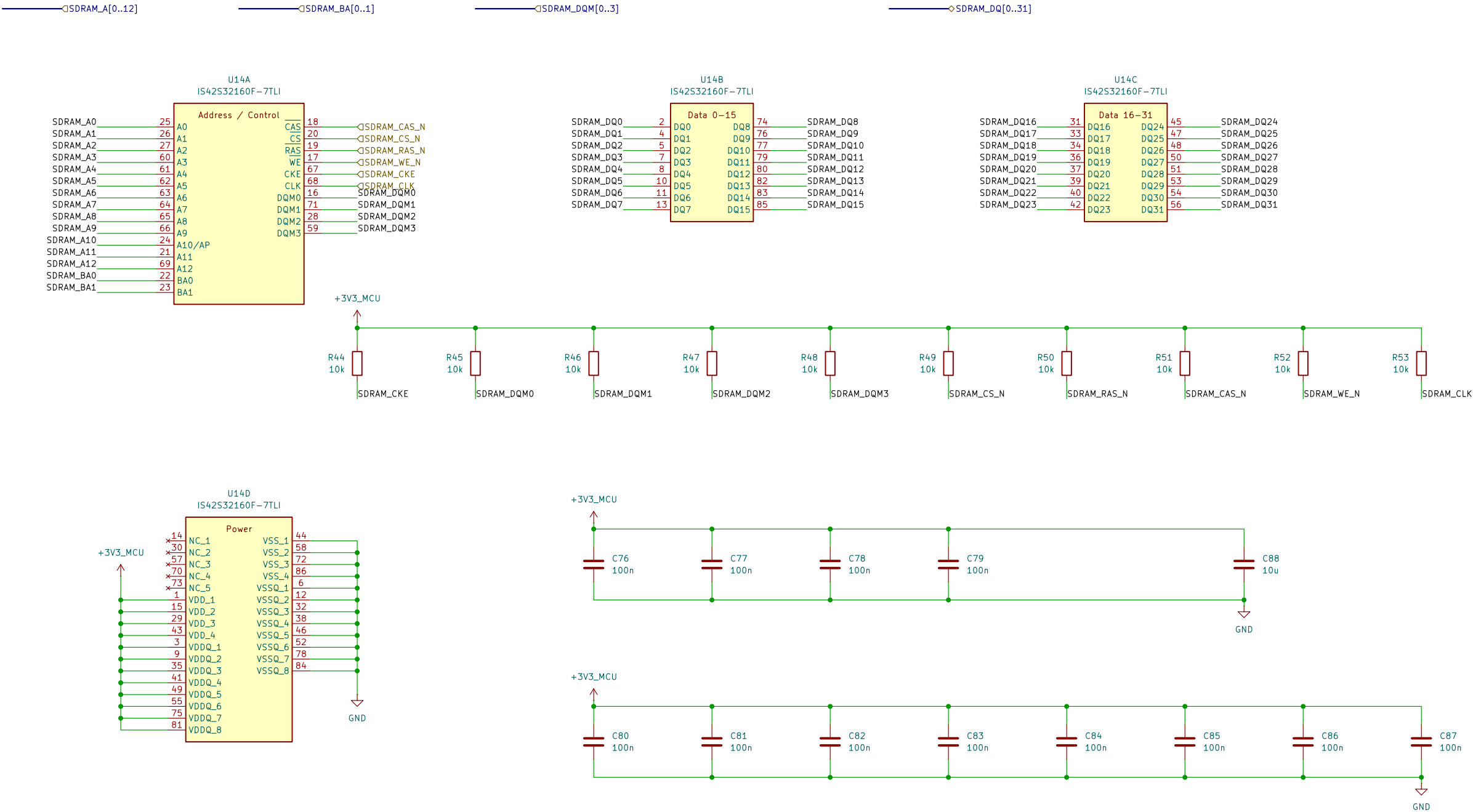
SENS-010 discharge revision: Rmax11.4211ohm; return/injection<=2.5mA.
Includes1.5mA key return +1mA ether backfeed: Vinf=0.028552750V.
t=1m*11.4211ohm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.398757ms total, below held75.794309ms screen.
Includes10ms command/clamp/discharge response and10ms below90mV.
Held budget:1.825mA control +0.817mA reverse;1uC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON source protection, loaded startup and fast faults remain open.
RST-002 / RADIO-019: release margin remains conditional on rail/path bounds.
Do not power up until reset coordination and source protection are complete.
Full assumptions: ../design/main_regulator_tps63806.md;
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

Sheet: /Main 3V3 digital supply/
File: main_3v3_supply.kicad_sch

Title: RA8P1 E-reader – main digital 3.3 V supply

Size: A3	Date: 2026-09-08	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.6	Id: 9/14	

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

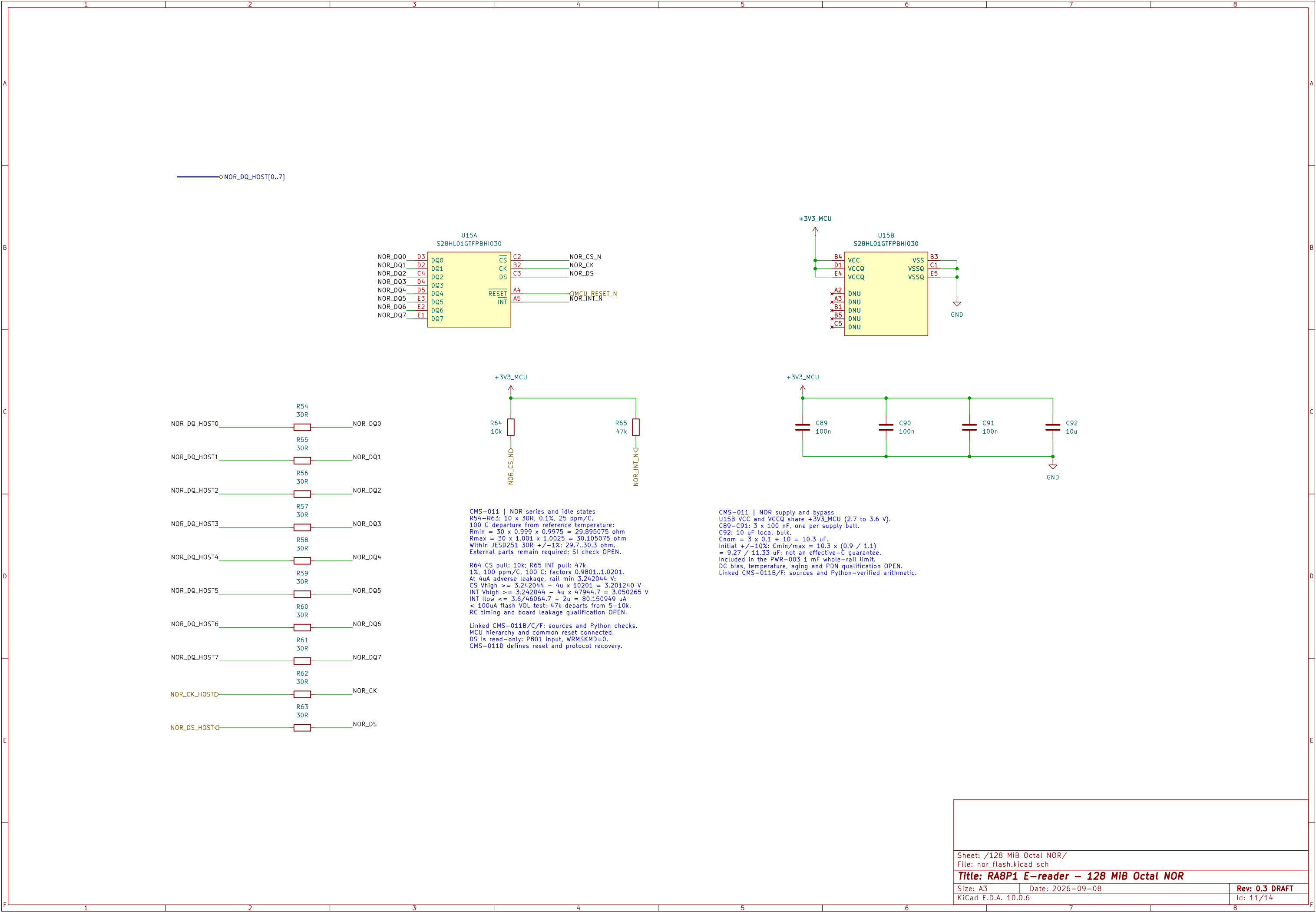
Size: A3

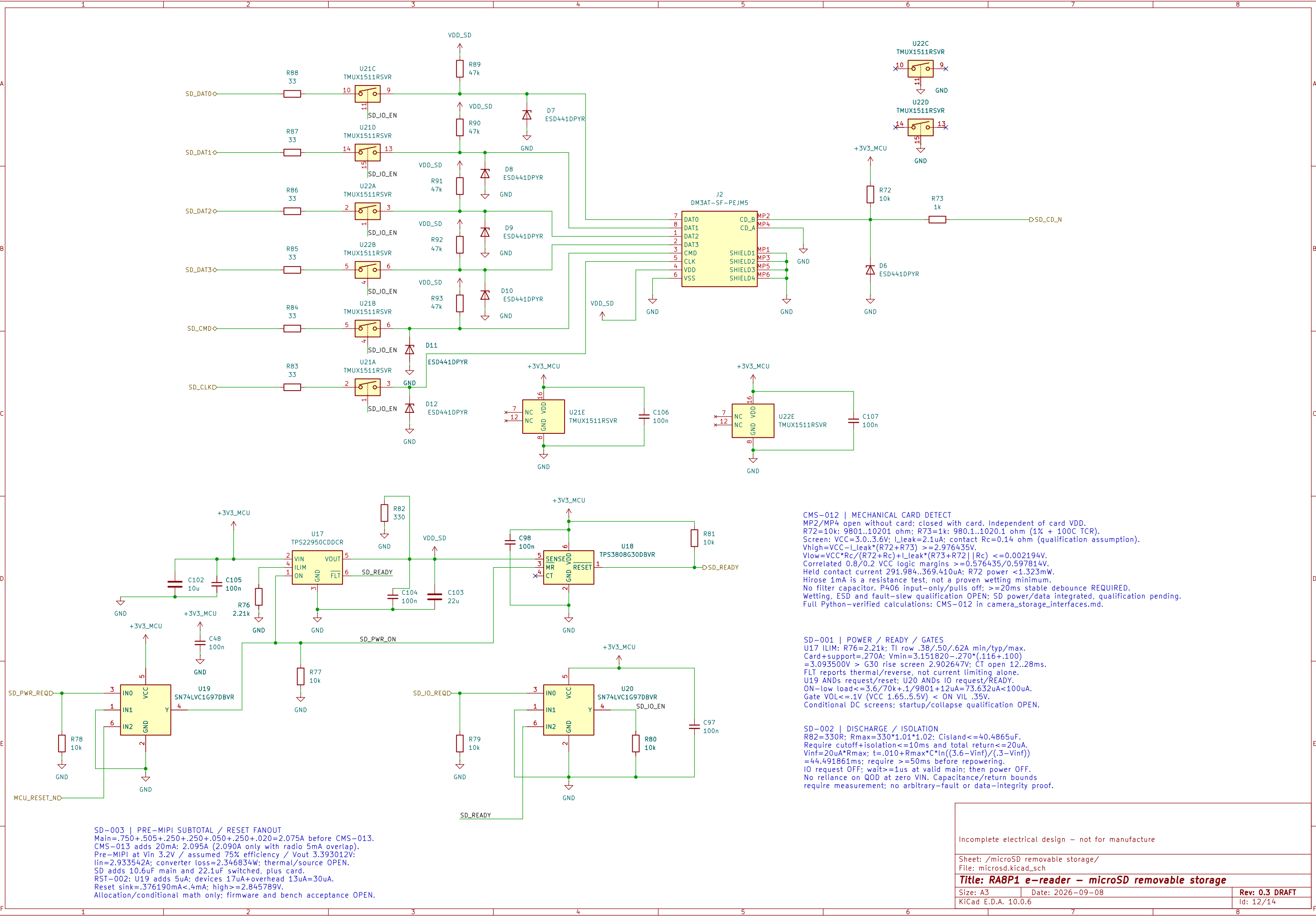
Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 10/14





CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k; 9801..10201 ohm; R73=1k; 9801..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; I_{Leak}=2.1uA; contact R_c=0.14 ohm (qualification assumption).
V_{high}=VCC-I_{Leak}*(R72+R73) >=2.976435V.
V_{low}=VCC*R_c/(R72+R_c)+I_{Leak}*(R73+R72)|R_c <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hiros 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; V_{min}=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < 0N VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; R_{max}=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff+isolation<=10ms and total return<=20uA.
V_{inf}=20uA*R_{max}; t=.010+R_{max}*C*ln((3.6-V_{inf})/(.3-V_{inf}))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
I_{in}=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

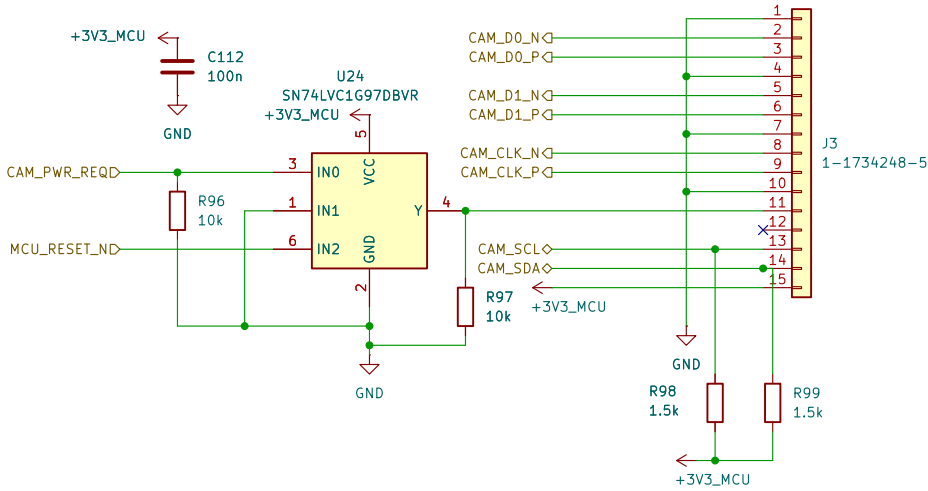
Incomplete electrical design – not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

Title: RA8P1 e-reader – microSD removable storage

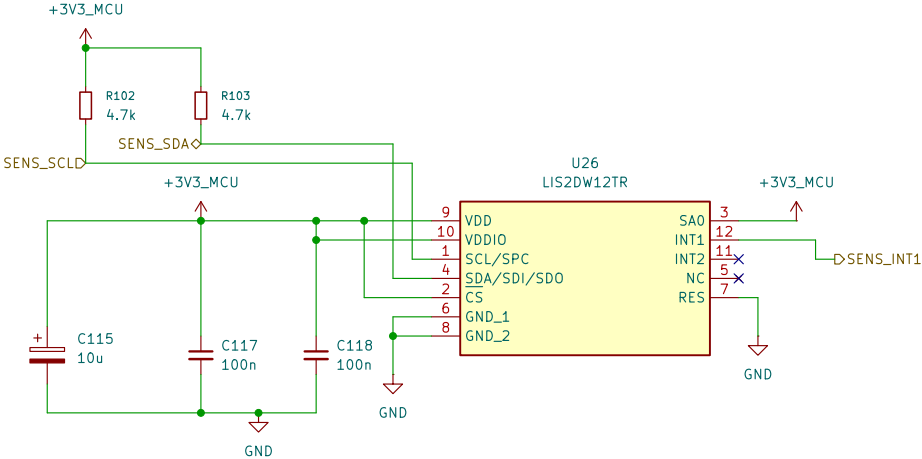
Size: A3 Date: 2026-09-08
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Rev: 0.3 DRAFT
Id: 12/14



CMS-014 / CMS-015: Pcam 5C, 2-lane CSI-2 to VIN0
U24: PWUP = CAM_PWR_REQ AND MCU_RESET_N.
PWUP low >=100ms; then high >=50ms before SCCB.
Qualification and second-camera/lighting work remain open.
See ../design/camera_storage_interfaces.md, CMS-014..016.

ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



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Rev:

Id: 14/14